

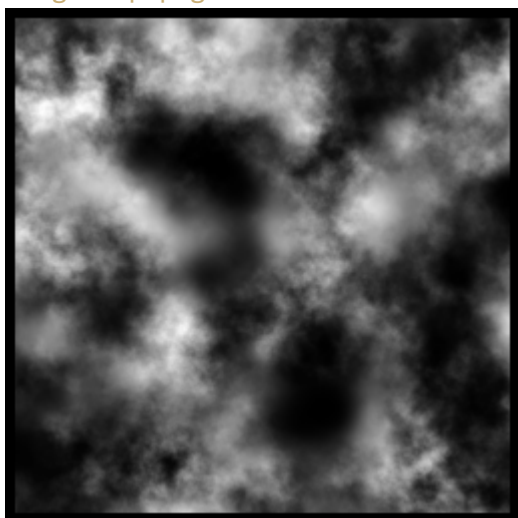
Zadanie 6 - Robert Mesek

heat_time_matlab.m

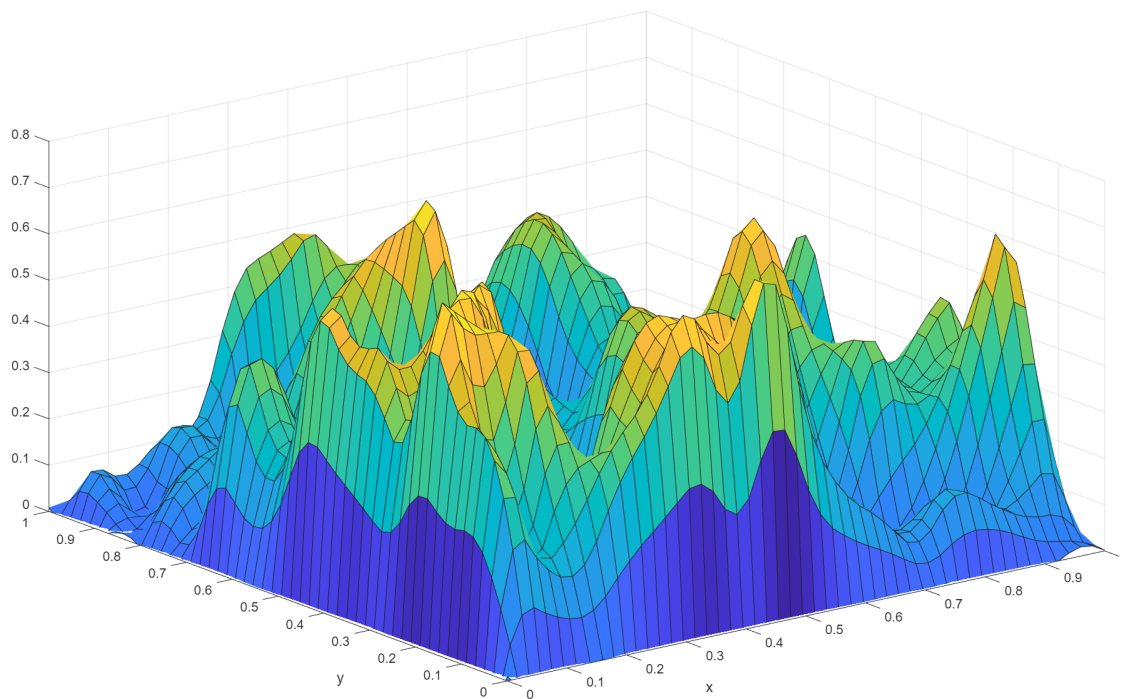
```
% Input data
knot = simple_knot(16, 2); % knot vector (16,2)
                                %[ 16 intervals and quadratic B-splines]
dt = 0.0001; % time step size
theta = 0; % scheme parameter (0 - explicit Euler, 1 - implicit
Euler, 1/2 - Crank-Nicolson)
K = 100; % number of time steps
% [...]
f = @(t, x) 0;
init_state = @(x) init_state_from_image(x, "heightmap.png");
% [...]
function y=init_state_from_image(x, filename)
    img = imread(filename);
    img = flipud(img);
    img = rgb2gray(img);
    img = im2double(img) * 0.8;
    [h, w] = size(img);

    col = round(x(1) * (w - 1)) + 1;
    row = round((1 - x(2)) * (h - 1)) + 1;
    y = img(row, col);
end
% [...]
```

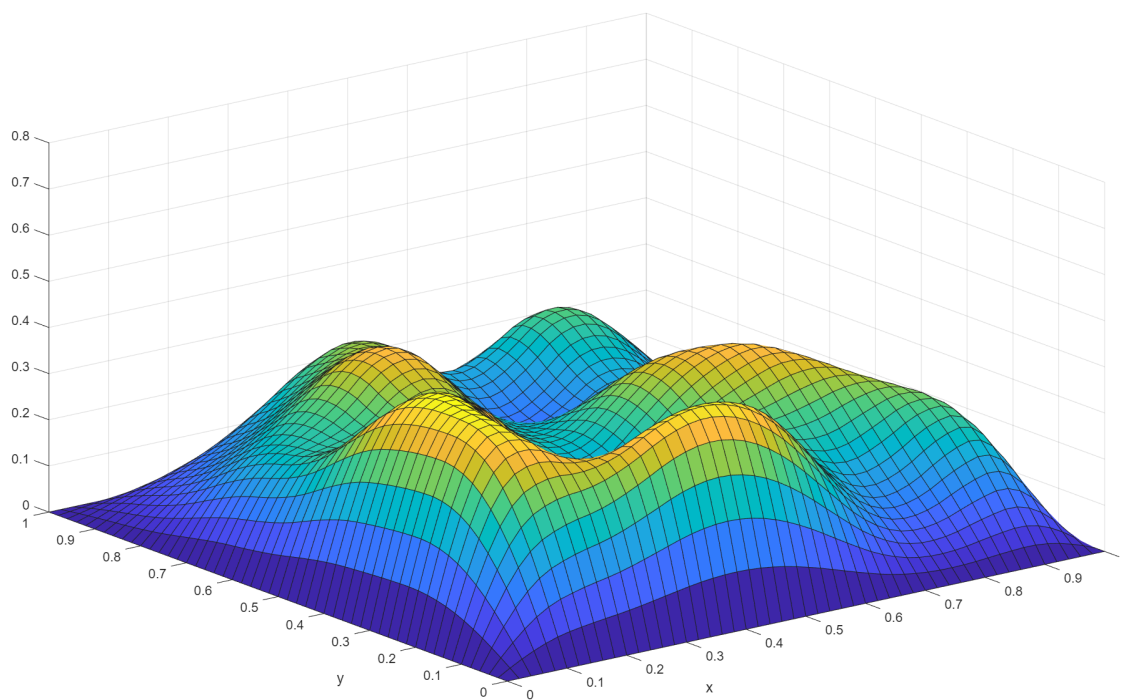
heightmap.png



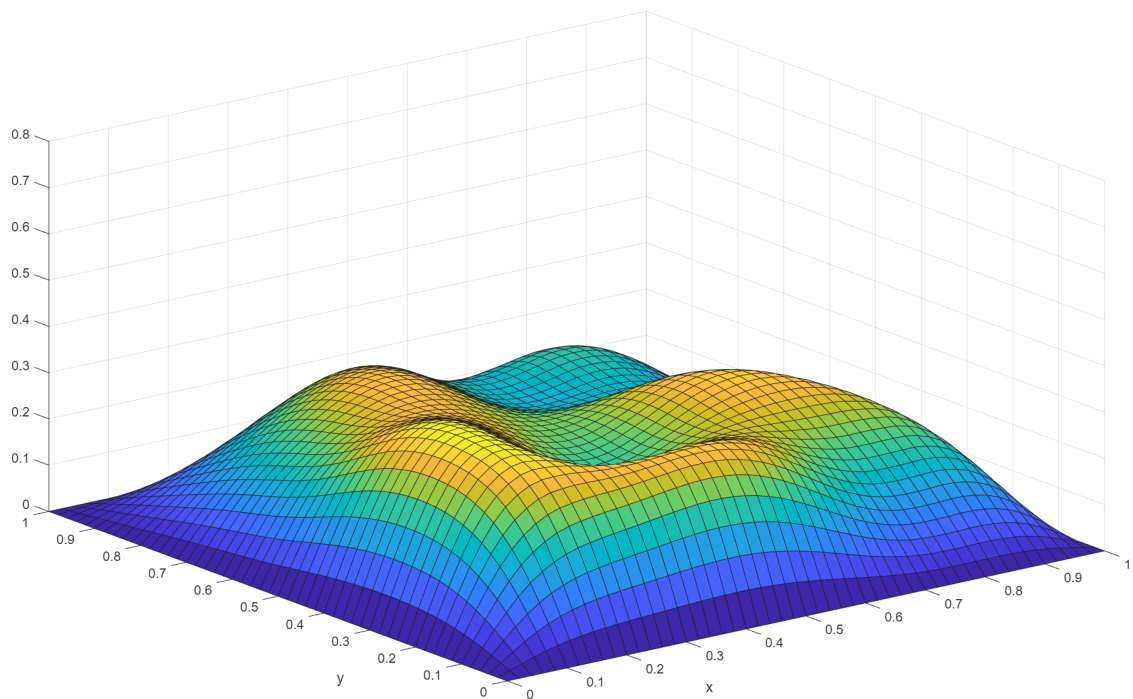
out/frame000.png



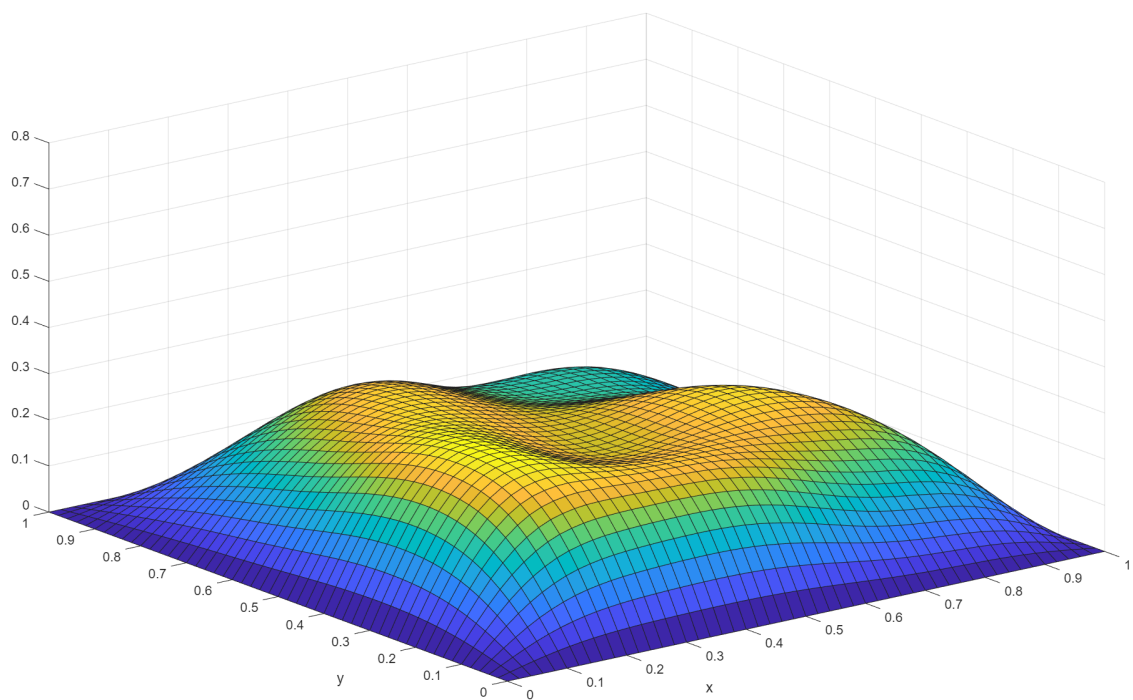
out/frame025.png



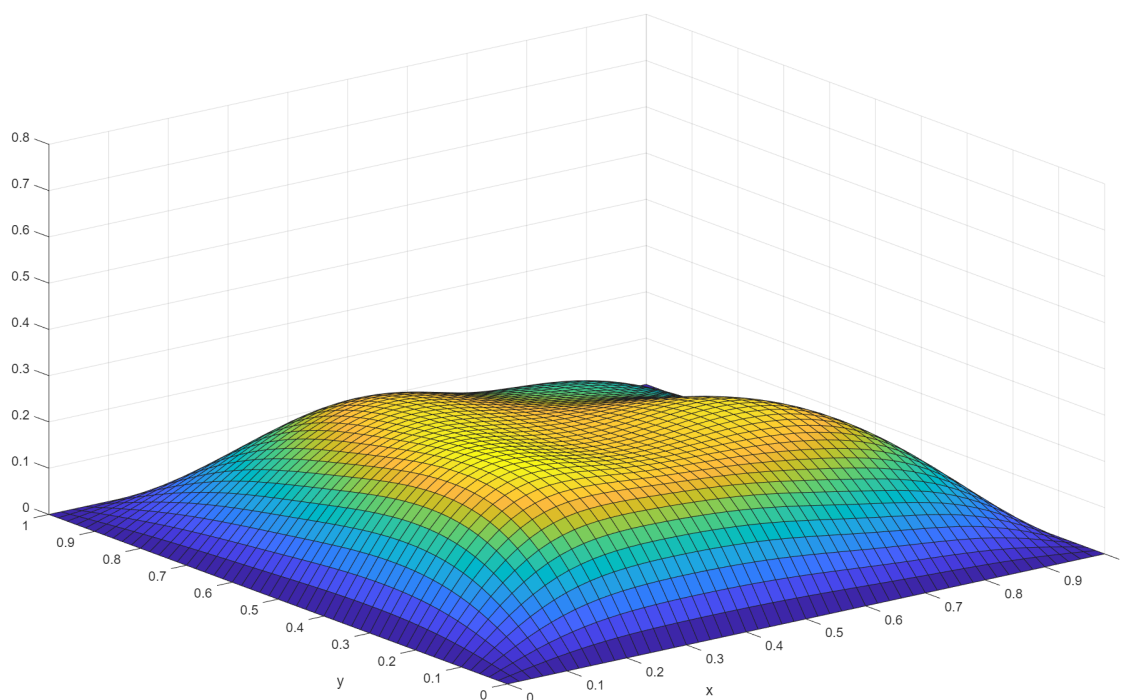
out/frame050.png



out/frame075.png



out/frame100.png



Animacja została stworzona z wykorzystaniem strony ezgif.com